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ReScene: A Hybrid Semantic and Structural Framework for Revisit-Based Video Forgery Detection

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CERTIFICATE

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We Hereby Certify that the Work Which is Being Presented in the Major Project Phase I Entitled "ReScene: A Hybrid Semantic and Structural Framework for Revisit-Based Video Forgery Detection" in Partial Fulfillment of the Requirements For the Award of Degree of Bachelor of Technology and Submitted in the Department of CSE(Artificial Intelligence and Machine Learning), B V Raju Institute of Technology, Narsapur, is an Authentic Record of Our Own Work Carried Out During the Period From June 2026 to October 2026, Under the Supervision of Dr. B VeeraShekarReddy, Assistant Professor and Ms. B Priyanka, Assistant Professor.

The Work Presented in this Major Project Phase I Report Has Not Been Submitted By Us For the Award of Any Other Degree of This or Any Other Institute/University.

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ABSTRACT

With the development of more sophisticated video editing and AI-generation tools, it becomes harder and harder to detect the authenticity of video content. The existing forgery detection approaches are mainly concentrating on low-level visual artifact or pixel-level inconsistency, and thus ignore semantic manipulation in the content of a real scene. We present ReScene, an approach to detect video forgery in revisit scenario by taking advantage of scene-level semantic reasoning and visual structure verification.

Our ReScene operates on the reference video and revisit video in two different yet compatible pipelines. Semantic pipeline involves object detection via YOLOv8 on representative frames, and each representative frame is then converted into a scene graph where the objects are nodes and the relations between them (such as left-of, right-of, above, below, near) are the edges. MemoryGCN will analyze the scene graph and detect semantic inconsistencies caused by object insertion/deletion/replacement/motion/relation changes. At the same time, structural verification pipeline is performed on the corresponding stable frames independently of object detection: SSIM will localize the areas with structural inconsistencies, and ORB will match local features of these areas.

The confidence measures of the semantic and structure modules are combined using an adaptive confidence fusion approach to give the tampering confidence measure, which points out suspicious components within the scene. ReScene integrates object-level semantic understanding, persistent scene memory, inconsistency detection through graph theory, and structure validation to give complementary information on the authenticity of revisit video scenes and manipulations at the object level.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AR	Augmented Reality
BA	Bundle Adjustment
CNN	Convolutional Neural Network
CSV	Comma-Separated Values
DETR	Detection Transformer
DSO	Direct Sparse Odometry
GPU	Graphics Processing Unit
IMU	Inertial Measurement Unit
JSON	JavaScript Object Notation
LiDAR	Light Detection and Ranging
MVS	Multi-View Stereo
NeRF	Neural Radiance Fields
ORB	Oriented FAST and Rotated BRIEF
PnP	Perspective-n-Point
RANSAC	Random Sample Consensus
RGB	Red Green Blue
RGB-D	Red Green Blue with Depth
SfM	Structure-from-Motion
SIFT	Scale-Invariant Feature Transform
SLAM	Simultaneous Localization and Mapping
TSDF	Truncated Signed Distance Function
ViT	Vision Transformer
YOLO	You Only Look Once

CHAPTER 1

Introduction

1.1 Background Of The Study

The rapid development of technologies such as artificial intelligence, deep learning, and multimedia editing has led to increased ease of video manipulation. The current editing technology allows the addition, deletion, replacement, and repositioning of objects without compromising the final result. This poses serious challenges for the authentication of videos used as evidence in applications ranging from surveillance, journalism, security audit, criminal investigations, insurance industry, to social media.

Detection of video forgery is more challenging compared to that of images due to the two-dimensional nature of video information. It is necessary for the forgery to be visually consistent over multiple frames to remain undetected. Therefore, traditional passive forensic approaches to video forgery detection rely on the low-level features including sensor noise, inconsistencies in compression, artifacts of color interpolation, temporal noise, and motion discontinuities. Deep learning techniques have enabled video forgery detection to become more sophisticated through learning representations from both real and forged videos. CNN, attention, and transformer models can detect inconsistencies based on the appearance and time. Nevertheless, a major limitation of existing techniques is that they concentrate mainly on visual evidence in terms of frames and pixels but not in terms of semantics.

The proposed ReScene framework addresses this problem by combining semantic scene understanding with structural visual verification. The semantic pathway constructs scene graphs using YOLOv8 and maintains a Scene Memory containing object appearances, relationship frequencies, and contextual co-occurrences. A MemoryGCN then analyzes this accumulated information to identify semantic inconsistencies.

1.2 Revisit-Based Video Forgery Detection

Revisit-based forgery detection in videos is concerned with finding discrepancies between various observations of the same physical scene recorded at different points in time. In contrast to the usual video forgery detection that could consider a single image or a video separately, the revisit-based scenario offers an extra piece of evidence, namely, the earlier observation of the scene. The manipulated object may look visually real but at the same time contradict the earlier observation. As a result, the proposed framework addresses the problem of revisit-based video forgery detection through multi-level consistency analysis, where semantic information indicates the changes that occur, while structural information aids in analyzing the consistency of the observed visual parts based on scene observations. This approach serves as the basis for the proposed method of detecting scene and object manipulation within videos, namely ReScene.

1.3 Semantic Object Detection

Semantic object detection is one of the vital steps in ReScene since it helps to create a meaningful representation of the visual content of the video scene. Instead of considering a video as a set of pixels, object detection finds specific objects and provides semantic classes for them. That is how it is possible to evaluate changes in objects incomparison with the previous observation of the same scene.For each frame in the input video sequence, the model of ReScene detects objects using YOLOv8. For each detected object, the class name and spatial coordinates are extracted from the output of object detection. The semantic object detection step provides the necessary semantic evidence of objects which are used to construct a scene representation and compare objects of reference and query observations. Object detection step is especially important for the revisit-based forgery detection scenario since it can happen that a manipulation in the scene changes its composition.

1.4 Problem Statement

Most existing methods for video forgery detection are based on either low-level visual cues or scene understanding based on semantics. Appearance and pixel-based techniques can detect visual distortions, but they might not be able to determine whether the relationships of objects have been preserved between multiple views of the same scene. On the other hand, scene-graph-based techniques can infer object relationships but depend on the abilities of the object detector.

In revisit-based video scenarios, a previously seen environment can be modified in terms of adding an object, deleting an object, replacing an object, displacing an object, or changing the relation between the objects.

1.5 Motivation of the project

The growing capability to produce forged videos through state-of-the-art video editing and generation has enabled the creation of manipulated videos that seem visually realistic. The existing methods of video forgery detection tend to rely on low-level visual clues or inconsistencies present inside the video itself. In this case, however, such evidences might not be enough as long as the manipulated part remains realistic and lacks any visual inconsistencies. Another context in which video forgery detection can be carried out is a revisit-based setting where the same physical scene is captured multiple times at different times. In this case, the prior observation can act as the reference in the process of analyzing the latter one. Any changes in the presence, identity, position, and relations between objects can thus serve as evidence of forgery. ReScene aims at utilizing this additional information through the use of semantic and structural evidences. While the former allows identifying inconsistencies related to objects and their relations, the latter gives another evidence related to the changes in scene regions.

1.6 Objectives of the project

A Hybrid Semantic and Structural Framework for Revisit-Based Video Forgery Detection is focused on the creation of such a framework that can recognize inconsistencies between reference and query videos depicting the same scene using both semantic scene understanding and structural visual verification methods.

- To build a video forgery detection framework that is revisit-based and performs comparisons among different observations of the same physical environment to find the possibility of any manipulations.
- To recognize and represent objects semantically in video frames through an object detection model, giving information about the objects found in every scene.
- To create scene level semantic representation of the objects and the spatial relations between them to be used for comparing the reference and query observations.
- To find semantic inconsistencies like object insertion, object deletion, object substitution, object displacement, and modification of spatial relations between revisited scenes.
- To integrate the structural visual verification process to be able to analyze the corresponding regions and give evidence for the scene changes.
- To integrate both of the semantic and structural evidences in a single hybrid framework for evaluating the consistency of the differences observed as potential video forgery.
- To offer the evidence of the manipulation with the help of relating the suspect changes to objects and regions within the scene.

CHAPTER 2

Review Of Literature

2.1 Related Work

The FaceForensics++ dataset, developed by Rössler et al. [1], quickly turned out to be one of the most popular datasets used to test the effectiveness of facial manipulation detection systems. However, as its name suggests, the FaceForensics++ dataset is limited to the analysis of faces – there is no need to check whether the rest of the scene or some objects within it have been manipulated. In contrast to the former, the ReScene dataset serves completely different purposes. It aims to detect scene and object inconsistencies in videos of revisits to the same environment. Thus, one may check for object insertions, deletions, and swaps using the ReScene dataset.

The DFDC preview dataset, which has been introduced by Dolhansky et al. [2] aimed at face swap detection on a large scale and various lighting and compression cases. Similar to FaceForensics++, this method is focused on the face: swapping, identity manipulation but not on whether the light source from the background was moved or a chair has been deleted from the scene. And again, here comes the ReScene, which aims to make sure that the whole scene remains consistent.

Another more sophisticated method is DeeperForensics-1.0 [3], which performs face swap operations to simulate noise and transmission errors. Its aim is to analyze the resistance of forgery detectors in the case of visual distortions, but not to detect any environmental modifications. However, ReScene pays attention to the detection of Celeb-DF [4] by Li et al. cares a lot about DeepFakes of good quality, smoothing out their glitches, such as color inconsistencies and contours of faces. The algorithm is designed to spot even the tiniest facial details, yet it does not mind posters being missing or sofas disappearing from the background. environmental inconsistencies, not just subtle facial cues.

ForgeryNet was developed by He et al. [5] to be a large dataset that is highly annotated both at spatial and video level with regard to manipulations involving faces, such as swapping, reenactment, and editing. This data does not care whether some additional objects were inserted in the background or a plant was changed. This is precisely what ReScene is about.

The WildDeepfake dataset [6] is created by Zi et al. The data is collected from the Internet, meaning that the video has actual artifacts of compression and capturing. However, all the attention is paid to the faces; no concern about the surroundings or objects is shown here. In contrast, ReScene works differently.

While KoDF [7], which is created by Kwon et al., provides diversity in the demographics of deepfake databases by having information about Korean faces, expressions, and speech, it focuses on expressions and head pose just like the others and not on the scene. ReScene focuses on appearance/disappearance of objects, and this is something that facial databases lack.

Approaches to detect deep fakes have been classified in terms of algorithm robustness by Wang et al. [8]. Their research remains limited to faces, reality and fake, and lacks spatial relationship between the objects in the scene, which is precisely where ReScene helps.

Pei et al. [9] discuss the present state-of-the-art of facial deepfake synthesis and detection, exploring the recent architectural innovations and defense measures. This review sticks to facial domain, ignoring issues related to physical scene integrity. If you want to know if some plants were swapped or if a picture frame was changed, you should use ReScene.

DeepfakeBench [10] by Yan et al. establishes a benchmark space to evaluate face manipulation detection algorithms in a unified way using a standardized protocol and dataset focused on faces and not scenes or objects in the image. This is precisely the focus area of ReScene, i.e., object consistency in repeat visits.

Face X-ray [11] by Li et al. analyzes the splicing artifact of the face region and its border with respect to the background frame. This technique focuses exclusively on the face border and not on changes to any object in the frame outside the face. This is precisely what ReScene handles.

Self-blended Images (SBI) [12] is a framework by Shiohara and Yamasaki that trains detectors to identify common synthesis artifacts by blending real faces. Again, this approach solely considers face borders and blending and does not consider context and scene objects across repeat visits like ReScene.

Frequency-Aware Framework [13] by Qian et al. extracts high-frequency artifacts produced by generative face models and identifies the anomalies in the frequency domain present in the face region. This technique ignores object semantics and their change across repeat visits which is considered by ReScene. In [14], Zhao et al. propose multi-attention models to study facial landmark manipulation artifacts. The emphasis on facial analysis makes them miss the broader scope of room-level manipulations. In contrast, the proposed method, ReScene, is able to detect unusual manipulations of space.

The local relation learning mechanism of Chen et al. [15] is used to detect inconsistencies between patches on the face. It is excellent in detecting intra-face relations but does not care much about the coffee mug in the background. This is precisely what the proposed technique, ReScene, excels at.

The representative forgery mining technique proposed by Wang and Deng [16] is used to detect local facial artifacts in the manipulated regions. They focus on finding cues in facial images and ignore the changes at the object level in the scenes. This is exactly the opposite of ReScene.

The self-consistency framework of Zhao et al. [17] checks for the consistency of the identity and textures of patches on the face. There is no discussion about the changes in the room's layout and object consistency.

SLADD [18] from Chen et al. uses self-supervised adversarial learning to effectively detect forgery faces that focus only on the face and not external scene reasoning. If you need environmental reasoning, check out ReScene.

UIA-ViT by Zhuang et al. [19] utilizes vision transformers for detecting inconsistencies in facial regions without supervision. This approach is tailored only for face analysis and fails to consider objects' movements in the background. This problem is solved by ReScene.

AltFreezing [20] by Wang et al. divides training into spatial and temporal modules to detect fake faces in videos. It efficiently detects dynamics of faces, but fails to notice changes of background objects like a sudden disappearance of a chair in a video. That's exactly what ReScene does.

SPatial-Phase Shallow Learning (SPSL) of Liu et al. [21] uses spatial and phase information for the detection of up-sampled images. Again, the focus remains on facial artifacts and not large scale scene or object level alterations which are the forte of ReScene.

Luo et al.

[22] relies heavily on high-frequency features for the detection of face forgeries, but again, the technique is restricted to faces only and not the object-level manipulations within a scene.

Dong et al.

[23] try to draw attention to the manipulated regions in faces in an efficient manner but do not consider any contextual aspect of the scene. This is precisely what ReScene takes into account.

The identity-driven technique of Gu et al.

[24] attempts to detect face swaps by separating appearance and identity information – a narrow approach ignoring objects and their motion in the background scene altogether.

Wang et al. [25] rely on latent space augmentation to increase the robustness of forgery detection across facial domains. Scene-level object swapping is not within the scope of their research; ReScene is designed for it.

Nirkin et al.'s context-discrepancy approach [26] examines the seamlessness between the face and its surrounding area – the hair, neck, and body parts. However, this is its limitation; it cannot detect object-level scene changes that fall under the focus of ReScene.

Zaidi et al. [27] offer a review of Vision Transformers for facial forgery. Their review is concentrated on faces, not multi-view scene layouts, unlike ReScene.

Khormali and Yuan's graph transformer method [28] concentrates on facial landmark geometry and structure anomalies, not on how separate objects interact during scene visits.

FSBI of Liu et al. [29] improves forgery detection by means of training on high-frequency, self-blended facial boundaries – strictly facial, not object or scene level.

However, while all the aforementioned works deal with digital forgery detection, deepfakes, and even 3D change analysis, ReScene is different in its attention to the physical manipulation of objects in a scene and corresponding changes benchmarking upon revisit.

2.2 Foundations Of Video Forgery Detection

First, one should explore the concept of video forgery detection through the exploration of how the authenticity of visual content can be evaluated based on information contained in a video. Video forgery detection tries to detect any possible manipulations that were done to a video using visual artifacts, temporal inconsistencies, structural variations, and other traces left in the video due to the editing process. In contrast to image forgery detection, the analysis of videos requires the consideration of the connections between consecutive frames since the manipulation of a video can affect both the appearance of a certain frame and temporal consistency of the whole video. Therefore, the earlier and well-established forensic techniques analyzed the features like compression artifacts, pixel level discrepancies, texture variations, frequency domain information, and temporal changes in order to evaluate the authenticity of the visual content.

With the emergence of deep learning, convolutional neural networks, as well as other types of learned representations, were applied extensively for manipulation detection in visuals. Such methods are able to discover discriminatory features directly from the authentic and forged data, allowing detection of subtle artifacts, which may be hard to define in advance. Some of the relevant approaches include Face X-Ray [11], F3-Net [14], Multi-Attentional Deepfake Detection [15], and self-supervised artifact-discrepancy learning [13]. Some approaches focus on self-consistency of local regions as well as dependencies between them, demonstrating that inconsistencies between related visual parts could serve as valuable forensic clues [16], [18]. However, the vast majority of these methods concentrate only on the face manipulations or low-level visual artifacts, but not on the semantics of a physical scene.

Another important basis for forgery detection in video is temporal analysis, since a video is composed of information scattered across its frame sequence. Some forms of tampering may cause discontinuity in appearance, motion, frame sequence, and frame repetition, which can be detected through temporal analysis. Thus, there are various studies on temporal dynamics, frame duplication, scene change aware analysis, optical flow, inconsistencies in motion blur, and inter-frame residual [32], [36], [38]–[50]. It is shown that relationship analysis among video frames can reveal evidences that cannot be derived from an individual image. But temporal consistency among frames is not the same thing as semantic consistency between two observations of the same physical scene. Even if an object has been tampered with in some way in the scene visitation process, the video will still be temporally consistent.

The use of semantically and relationally encoded information offers another way of improving forgery detection for videos. Techniques such as Local Relation Learning [16], Learning Self-Consistency [18], Semantic Contextualization of Face Forgery [22], and AUNet [23] have shown that information regarding the relationship between

meaningful visual parts can offer further insight into the manipulation process. Rather than treat each part separately, these techniques take into consideration the relationship between the parts. This form of relational reasoning becomes very important for scene-level forgery since the realism of a physical setting does not only rely on the individual objects in the scene but also on their spatial relationship with one another. However, the current state-of-the-art solutions discussed above in the literature all focus on facial forgery detection.

A complementary source of evidence is provided by structural verification. Techniques such as local features, similarities, residuals, and others can find visual regions that are different across the two observations. Frame duplication and similarity-based techniques, for example, show that the correspondence between visual observations can be used to find suspect differences [38]–[42]. In some cases, however, differences in structure alone do not guarantee that the differences indicate forgery. Differences in lighting, perspective, object appearance, or even differences in the way images were recorded can cause differences in visual observations. In these cases, structure needs to be complemented by semantic knowledge about objects and relations between them.

Even though there is a solid basis from previous research that utilizes visual, temporal and structural inconsistencies, many of the approaches still focus on the issue of detecting artifacts in a single video or even face. However, they fail to deal with the problem of checking if the composition of the scene semantically stays consistent when the same environment is revisited once again. This issue is the motivation behind revisiting formulation in ReScene. This framework uses semantic object detection and relational scene understanding along with separate structural verification. The first one is responsible for recognizing inconsistency at the level of objects and relations between them, whereas the second branch relies on SSIM and ORB feature matchings and gives additional visual evidence. This way, both

types of evidence lay the groundwork for ReScene hybrid framework of revisit-based forgery detection.

2.3 Deep Learning-Based Video Forgery Detection

The progress in deep learning technologies revolutionized the development of video forgery detection systems through the replacement of forensic features handcrafted by humans with those learned from visual content directly. The convolutional neural network became one of the fundamental technologies used for such a purpose since they are capable of learning hierarchies of spatial features from forged and unaltered frames. The studies in deepfake detection have mostly focused on facial manipulations, in which visual artifacts resulting from the process of manipulation were learned by the model. The benchmarking datasets such as FaceForensics++, DFDC, Celeb-DF, DeeperForensics-1.0, ForgeryNet, WildDeepfake, and KoDF included many samples of forged videos [1]–[7].

With the evolution of deep learning-based techniques, efforts were made towards exploring more advanced representations for the detection of manipulation artifacts. Face X-Ray [11] considered artifacts resulting from the blending process at the boundary regions of the face where manipulation occurs, while F3-Net [14] considered the use of frequency features to reveal artifacts that could not be identified using spatial features. In addition, multiple attentions in Multi-Attentional Deepfake Detection [15] were introduced to highlight discriminative regions, and self-supervised techniques such as artifact-discrepancy learning [13] tried to uncover manipulation artifacts without completely relying on the handcrafted artifact templates.

Another research direction was devoted to relations and consistency across various regions of an image or a video. Local Relation Learning [16] focused on relations between regions of faces, and Learning Self-Consistency [18] relied on consistency information as a forensic cue. The Semantic Contextualization of Face Forgery [22] highlighted the role of contextual information in analysis of manipulated images, and

AUNet [23] was based on relations between facial action units in order to detect face forgery. This shows the shift from detection of isolated artifacts towards taking into account dependencies between semantically significant regions. Nevertheless, most of the above-mentioned works are developed with face content in mind and, thus, do not reflect the semantic structure of a real physical environment.

Deep learning techniques have also been extended from single frame analysis towards temporal video analysis. Techniques such as those studying temporal inconsistencies [32], temporal dynamics [36], recurrent convolutional techniques [37], and spatiotemporal inconsistency learning [46] try to identify the change that happens between two consecutive frames. There are other methods which study frame duplication and inter-frame similarity [38] - [42], and also optical flow, motion blur, and inter-frame residual methods [48] - [50] which employ motion and structural difference as a forensic indicator. These methods prove the significance of temporal data because any tampering might cause some inconsistency that becomes evident over time.

Despite these developments, deep learning-based video forgery detection remains strongly influenced by the characteristics of the training data and the manipulation types represented within it. A model trained mainly for the detection of manipulations in faces will acquire features that are manipulation-specific or identity-specific, and not necessarily scene-level features. The temporal or appearance model can detect discrepancies but without specifying which objects have been manipulated or whether the relations between them are preserved. The problem of generalization [61]–[67] underscores the difficulties of reliable detection despite changes in the manipulation technique, dataset, architecture, or visual conditions.

However, the approach for the detection of video forgeries with deep learning is strongly dependent on the dataset. A detector that mainly works on facial manipulations learns the specifics of manipulation or individual identity rather than scene-level information. Also, temporal

or appearance-based models detect the inconsistencies regardless of the identification of manipulated objects or the spatial consistency of objects. Generalization researches [61]–[67] additionally highlight the more general problem of the robustness of detection against changes in forgery method, dataset, architecture, and visual conditions.

Thus, deep learning plays a crucial role in the functioning of ReScene, but not as the exclusive source of forensic data. The approach of ReScene is to expand the scope of deep learning beyond the artifact-based detection framework to the semantic level by performing a scene-level consistency analysis. YOLOv8 model is utilized to get object-level semantics and spatial relationships between objects are encoded in a scene graph, which is stored in scene memory. Inconsistencies between the reference and the revisit observation regarding object existence and relationships are determined using graph-based reasoning. A standalone structural branch that relies on SSIM and ORB features enables ReScene to use both learned and structural evidences.

2.4 Semantic And Spatial Reasoning For Video Forgery Detection

With video forgery detection going past low-level visual artifacts analysis, researchers started looking into the possibility of finding extra clues through semantic information and connections between visual regions in videos. Rather than considering video as an aggregation of separate pixels or frames, semantic approaches seek to find entities and the way they interact. In real-life scene, objects do not exist independently from each other; the connection between them and their interactions give information about the scene. Thus, finding change in any entity, its positioning, or interaction with surrounding objects may help detect forgery differently from traditional pixel-based approach.

Relation information has been explored in a number of deepfake detection papers. Local Relation Learning [16] analyzes dependencies between different facial regions rather than analyzing each one

separately. In a similar fashion, Learning Self-Consistency [18] looks at the consistency between various regions of manipulated content. The way facial action units relate to each other is modeled by AUNet [23], while Semantic Contextualization of Face Forgery [22] takes into account semantic context in forgery detection process.

2.5 Structural And Temporal Consistency Analysis

The next relevant avenue of research in video forgery detection is the investigation of structure and temporal consistency of video frames. Whereas semantic methods seek to understand what objects there are in the scene and how they relate to each other, the structural methods try to discover whether the spatial and local configuration of a scene is consistent. It is especially important in video since although a manipulation does not create an artifact in one frame, it can create inconsistencies when two consecutive frames or even different views of the same scene are considered.

Temporal consistency has been widely studied in relation to finding such inconsistencies. Inconsistency detection based on temporal analysis [32] seeks to find variations in video frames and discover the presence of abnormal manipulations. There are also methods which employ temporal dynamics [36] and recurrent convolutions [37] to learn evolving patterns. Spatiotemporal consistency learning [46] and video level temporal attention [47] prove that inconsistent information scattered in multiple frames can be helpful for the forensic analysis.

The importance of the above limitation becomes more apparent in the case of revisits, whereby the query video and the reference video do not have to be recorded from the same perspective and under the same conditions. The same scene can have a different structure even if the meaning of the scene has not been altered. On the other hand, the introduction of any new object or the removal of some objects results in a structural difference which is rather localized and needs to be interpreted semantically.

2.6 Scene Graphs And Graph-Based Visual Reasoning

Semantic scene understanding by representing a visual scene as a graph structure represents yet another promising direction. In contrast to the typical approach where an image representation encodes only the appearance of a scene, scene graph encoding implies explicit representation of entities and relations between them. Entities can be encoded as nodes, and the relations (spatial, semantic, or interaction relations) can be encoded as edges. This kind of representation allows one to have a structured representation of a scene and reason about relations between objects, rather than treat each recognized object separately.

The use of relation information has been explored in forgery detection by methods modeling interdependencies between various visual parts. Local Relation Learning [16] studies relations between facial regions, Learning Self-Consistency [18] utilizes self-consistency of related regions as the source of forensic information, AUNet [23] explicitly models relations between facial action units, and Semantic Contextualization of Face Forgery [22] uses contextual semantic information in forgery detection. It demonstrates that relation information can be complementary to the appearance information about visual entities.

The graph-based representation also allows for reasoning across several observations. While the frame-based representation of a scene will be the scene observed at a certain point in time, the graph representation over several observations will capture repeated objects and relations. The above fact becomes very useful to the revisit problem, where the same physical environment will be observed from various views or even partially observable.

2.7 Scene Memory And Persistent Semantic Representation

An important constraint of traditional frame based video analysis is the independence of each observation. As a result, it is difficult to

accumulate information related to a scene through a long sequence of frames or observations. In contrast, for revisit forgery detection, the algorithm should be able to decide if the semantic composition of a real world environment stays same in the reference video and an observation. For this task, it needs a representation which is capable of accumulating information related to an object and its relations.

For instance, the scene memory may be created by accumulation of semantic observations from the video sequence. In such case, an observed object would contribute information related to its semantic category as well as spatial location and repeated observations would contribute information related to its existence and relations with neighboring objects. In place of replacing the previous observation when a new frame is being observed, the memory representation would enable the accumulation of information over many observations.

In the case of ReScene, the scene memory is represented in a graph-like manner. The objects are semantic nodes, and the spatial relations between objects create the edges between the nodes. Multiple views of the same object or spatial relation contribute to its accumulation. Therefore, the memory can keep the information about the expected configuration of the scene even if there are some gaps due to the incomplete observations in individual frames. For instance, the absence of an object due to changing viewpoints and occlusions can be compensated for by the previously observed data.

2.8 Graph Convolutional Networks for Semantic Inconsistency Detection

Having a scene represented as a graph, the second requirement that arises is the analysis of the relations between the objects forming a scene. Standard neural networks are designed specifically for working with grid-structured data such as images; on the other hand, scene graphs have an irregular topology, with varying numbers of nodes and relations between them. Graph Convolutional Networks (GCNs) give a

way to learn graph representations using propagation of information between adjacent nodes.

A graph is defined as $G = (V, E)$, where (V) represents the set of object nodes and (E) is the set of relations between them. The node can store information about the semantic properties and context of the respective object, while the edges represent the spatial relationships such as left-of, right-of, above, below, and near. Using such a representation, the model can take into account not only the characteristics of a particular object but also the context of neighboring objects. As a result, an object is not evaluated as a stand-alone entity, but rather in context of a particular scene structure.

The relational processing in GCNs is done using the feature propagation in graphs. For a graph consisting of the node features matrix X and adjacency matrix A, one of the common approaches to graph convolution is formulated by the equation

$$\mathrm{H}^{(l+1)} = \sigma\left(\hat{D}^{-\frac{1}{2}}\hat{A}\hat{D}^{-\frac{1}{2}}H^{(l)}W^{(l)}\right)$$

2.9 Structural Similarity and Local Feature-Based Verification

Structural verification provides a complementary approach to semantic reasoning by analyzing the visual content of corresponding observations directly. While semantic analysis determines whether the objects and relationships within a scene are consistent, structural analysis examines whether the visual organization of the scene has changed. This distinction is important because certain manipulations may alter the appearance of a region without necessarily changing the detected object categories or their spatial relationships.

The Structural Similarity Index (SSIM) is a commonly used measure for evaluating structural similarity between two images. Rather than comparing pixels independently, SSIM considers local luminance,

contrast, and structural information. For two image regions $x$and y, the structural similarity can be expressed as

$$\mathrm{SSIM}(x,y)=\frac{(2\mu_x\mu_y+C_1)(2\sigma_{xy}+C_2)}{(\mu_x^2+\mu_y^2+C_1)(\sigma_x^2+\sigma_y^2+C_2)}$$

where $\mu_x$and $\mu_y$are local means, σ_x^2 and σ_y^2 are local variances, σ_{xy} is the covariance of the two regions, and C_1 and C_2 are constants used for stabilization. Larger SSIM value indicates more structural similarity while smaller SSIM value indicates structural difference. ReScene algorithm makes use of SSIM for finding regions with low structural similarity between stable frames.

ORB (oriented FAST and rotated BRIEF) is utilized in ReScene in order to perform verification of detected regions using the local features. The idea behind ORB is extraction of the unique local keypoints and descriptors which can be compared in different frames. It gives additional information about the fact whether the structures that were observed in the reference frame can also be observed in the revisit frame. Thus, in the proposed method, the regions with low structural similarity are further checked by means of ORB feature matching.

2.10 Adaptive Evidence Fusion for Video Forgery Detection

However, using multiple forensic cues can solve the problem of using a single source of evidence. The use of semantic analysis will reveal any discrepancies with respect to the objects and their spatial relationships within the scene, while the structural analysis will reveal any visual discrepancies that are not captured in the semantic analysis. Therefore, the two types of forensic cues can complement each other in video forgery detection by revisiting videos.

In ReScene, the output of the semantic pathway is the normalized confidence level $S_{\text{sem}} \in [0,1]$ based on how much semantic discrepancy is found using the MemoryGCN. The semantic confidence is determined through changes in the objects and their spatial relationships in the scene memory. Higher semantic confidence

implies more discrepancies in the semantic configuration between the revisit scene and its corresponding reference.

Simultaneously, the structural verification path generates a confidence score $S_{\text{struct}} \in [0,1]$, which is generated from the comparison of the reference and query stable frame structures using SSIM and then further verified using the ORB features. The larger value shows more evidence of structural inconsistency between the observations. As the pathway works directly with the image data without detecting any objects, it could generate the evidence of visual modification without having this change reflected in the scene graph.

The combination of the two confidence scores takes place through the use of the adaptive decision fusion approach. In the particular formulation described, the tampering confidence is defined as

$$S_{\text{final}} = \alpha S_{\text{sem}} + (1 - \alpha) S_{\text{struct}}$$

where S_{sem} is the semantic inconsistency score, S_{struct} is the structural inconsistency score, and α controls the weight of the semantic pathway. Such an approach makes it possible to make use of the evidence from both pathways in the final decision making process.

2.11 Identified research gap

From the review of literature, it is evident that forgery detection techniques have evolved from the initial lower-level forensic approach to deep learning, temporal modeling, semantic reasoning, structural verification, and graph based approaches. However, all these techniques have been developed independently with regards to each aspect of video forgery. The current deep learning models consider either visual artifacts or temporal artifacts. In case of semantic approach, all the relationships between significant regions are considered. In structural model, visual inconsistencies are detected but it doesn't mean that the semantic relevance is established.

2.11.1 Low-Level and Appearance-Centric Detection

A substantial amount of work on forgery detection in video forgery literature focuses on the presence of visual artifacts in individual

frames or a sequence of frames. Appearance-based methods using CNNs, attention mechanisms, Fourier transforms, and transformers learn representations that can recognize traces of manipulation [11]–[15], [20], [25], [28], [29]. Temporal methods leverage inconsistencies between frames [32], [36], [43], [45]–[50]. Although such methods may offer powerful forensic clues, the main representation used by these methods is appearance-based.

An appearance-based representation does not include information about the semantic structure of the real-world scene. This implies that although a system can detect visual inconsistencies, it cannot determine the specific action performed, namely whether an object is added, deleted, substituted, moved, or incorrectly positioned.

2.11.2 Limited Scene-Level Relational Reasoning

The research findings on relational information show that relations between visual elements can serve as an effective clue in forensic studies. Local Relation Learning [16], Learning Self-Consistency [18], Semantic Contextualization of Face Forgery [22], and AUNet [23] utilize relations and/or contextual dependencies in detecting the image forgery. However, the above-mentioned techniques are mainly applied to facial images and local semantics.

In terms of a generic physical scene, the relation between different objects is considered as significant contextual clue. Objects cannot always be assessed separately since their location and interaction with the objects nearby are needed in order to understand the scene as a whole. Thus, the literature creates a gap in scene-level relational reasoning for revisit-based forgery detection problem.

2.11.3 Absence of Persistent Scene Memory

Traditional video manipulation techniques consider either frames or temporal sequences to be an observation of the current video. However, a revisit technique demands comparison between a past observation of a reference scene and another observation of the same environment. Variations in the viewing angle, lighting, camera

placement, and partial occlusion can make the frame-wise comparison less reliable.

The surveyed literature offers the basis for semantic consistency and relational analysis [16], [18], [22], [23], but not for the persistent memory model to remember object appearance, relation frequency, and contextual co-occurrences over the entire reference video. ReScene fills this gap with the Scene Memory Model that retains semantic consistency over observations and offers contextual information for the revisit analysis.

2.11.4 Separation Between Semantic and Structural Evidence

The various structural approaches such as frame similarity, duplication, optical flow, motion features, and inter-frame residual [38]–[50] provide some insight about the visual change. But the structural difference does not imply any semantic modification. In turn, semantic analysis cannot identify any manipulations, preserving object category and spatial arrangement but changing its visual appearance.

It is evident from the literature that both types of evidence should be analyzed together. ReScene solves this problem using the parallel semantic and structural paths. The former analyzes the scene graph and MemoryGCN. The latter analyzes SSIM and ORB for corresponding stable frames.

2.11.5 Limited Focus on Revisit-Based Scene Forgery

The other limitation is the problem formulation itself. The majority of works on deepfakes deal with the problem of detecting whether the video/image under examination has been manipulated, in particular, its facial images [1] – [30]. Other papers are concerned with detecting inconsistencies in time, duplication, motions and structural alterations in videos [31] - [50]. All these papers do not formulate forgery detection as a problem of reference versus revisit scene consistency.

However, ReScene sees the reference video as the source of expected scene semantics and checks the revisit observation against that

semantics. Changes to be detected in the scene under examination include additions and removal of objects, replacement, motion of objects, and spatial relationships modification. This problem formulation moves the task from manipulation detection to the scene integrity violation detection.

2.11.6 Positioning of ReScene

Taking into consideration the above shortcomings, ReScene can thus be considered a hybrid video forgery detection system based on revisits which bridges semantic scene understanding and independent structural verification. The semantic path generates scene graphs through detected objects and employs scene memory that is kept in MemoryGCN for context reasoning. Meanwhile, the structural path employs SSIM and ORB for visual change verification, while adaptive confidence fusion merges both types of evidences.

Hence, the research gap that ReScene tries to fill can be considered as follows: There has been no existing system which integrates semantic scene-level memory and structural visual verification for the detection of semantic and structural changes between reference and revisit videos.

2.12 Limitations

While there has been much advancement made in the area of detecting forged videos using the available methods, the research shows that there are certain limitations associated with such methods in the case of revisit-based scene forgery detection. It is important to note that the limitations in question do not relate to any flaws of the methods as applied to their intended purposes.

2.12.1 Limitations of Appearance-Based Deep Learning Approaches

DL models that utilize CNNs, attention, frequency and transformers have been shown to be capable of learning manipulation-sensitive visual features [11]–[15], [20], [25], [28], [29]. Nevertheless, such DL methods mainly learn visual appearance and manipulation-related visual cues.

These representations do not carry information about the semantic structure of the physical environment. Thus, they might recognize changes in the region but do not represent whether an object is added, deleted, substituted or moved. Moreover, the performance of such representations depends on the type of manipulations and data distribution encountered during the training. Such issues are also addressed in generalization studies [61]–[67].

This issue motivated using a semantic pathway for ReScene in which the objects and their spatial arrangement would be represented instead of pixel-level learned features only.

2.12.2 Limitations of Facial Forgery Detection Approaches

The review of relevant literature reveals a number of works dedicated to the problem of facial forgery detection. Methods of FaceForensics++, DFDC, Celeb-DF, DeeperForensics-1.0, WildDeepfake, KoDF, among others, have been developed for investigating the problem of manipulated video [1]–[10]. There are other methods that analyze facial regions, action units, lip synchronization, and the relationship of the face context [16], [19], [22], [23].

On the other hand, facial representations cannot be directly applied to the environment, which may include such objects as pieces of furniture, electronics, containers, etc. Relationships of a physical environment differ from facial region dependencies. This is why, ReScene applies semantic reasoning to objects in a physical environment and their relationships.

2.12.3 Limitations of Temporal Forgery Detection Approaches

The temporal approaches explore the temporal inconsistencies present throughout the frames and have looked into temporal dynamics, temporal repetitions, frame duplication, scene change, optical flow, motion blur, and inter-frame residuals [32], [36]-[50]. These techniques will be very helpful to detect any anomaly which may show up over time.

But temporal consistency does not necessarily indicate the semantic state of the physical world. The video may still be temporally

consistent even after the deliberate addition or deletion of an object from the video scene, as long as the modification is done consistently over the frames that were manipulated. On the other hand, normal camera motion or shift in perspective may cause discrepancies without pointing towards any forgery.

This issue is solved by ReScene using semantic consistency analysis of the reference scene and revisit scene.

2.12.4 Limitations of Structural and Similarity-Based Approaches

On the other hand, structural comparison-based methods help to find any disparity between two corresponding visual observations whereas frame similarity and frame duplication techniques prove to be useful in case of visual tampering detection [38]-[42]. SSIM and local feature correspondence can be used in the same way to find out where there is a disparity in structural information. The biggest drawback associated with structural differences is that structural difference does not necessarily imply a semantic difference. Conversely, a manipulation may preserve much of the overall structure while changing the semantic identity or position of a particular object. Therefore, structural evidence requires contextual interpretation.ReScene uses SSIM and ORB as an independent verification pathwayrather than treating structural difference alone as definitive evidence.

2.12.5 Limitations of Semantic and Relational Approaches

Techniques that include both the semantic and relational contexts show that relationships among relevant regions could enhance forgery detection techniques [16], [18], [22], [23]. Yet, the examined techniques are primarily concerned with regions related to the face or other local semantic dependencies.

In the case of a real-life scene, there is a need to make reasoning about several object classes and their relationships. In addition, even a single frame would contain only a partial observation of the scene. Without the ability to keep any persistent memory, the system will be

unable to naturally collect knowledge about various objects and relationships.

ReScene solves this problem via its Scene Memory.

2.12.6 Limitations of Conventional Graph-Based Analysis

Graph representation is an appropriate way of representing relationships among objects and GCN can facilitate information propagation among connected vertices. However, a graph created from a single frame depicts the information of the scene in just that frame alone.

In case of forgery detection based on revisits, the requirement is that of information which exists not only in one but many frames. Transient absence of object, partial object visibility, and change in viewpoints can result in an incomplete representation of the scene. Hence, ReScene uses Scene Memory and MemoryGCN on top of graph representation.

CHAPTER 3

METHODOLOGY

3.1 Software Requirements

Operating System Windows / Linux / macOS

Language Python 3.8 or above

Environment Google Colab or local Python environment

FrameWork COLMAP (for Structure-from-Motion)

3.2 Required python libraries and frameworks

Python Libraries

- NumPy (numerical computation)
- OpenCV (frame extraction and processing)
- Pandas (data handling and CSV storage)
- Matplotlib (visualization)
- Ultralytics YOLOv8
- PyTorch (deep learning backend)
- SciPy (mathematical operations)
- JSON (semantic memory storage)

3.3 Tools and technologies

The presented ReScene framework utilizes different methods of computer vision, deep learning, graph-based learning, and image processing in order to detect video forgery both semantically and structurally. All the mentioned instruments are selected depending on their functionality, compatibility, reproducibility, and integration into open-source environments. All of the instruments are used to facilitate the stages of the framework, which include video and image processing, object detection, creation of the scene graph, graph-based semantic reasoning, numerical computation, and visualization of the experimental results. The Python programming language is the basis of the implementation of the instrumentations, while PyTorch and PyTorch Geometric serve as the deep-learning and graph neural network environments, respectively. Semantic object detection is

performed with the help of YOLOv8n, while video processing and structural analysis is performed with the help of OpenCV. NumPy provides the necessary functionality of numerical computation, while Matplotlib allows visualizing experimental results. The integration of the above-listed instruments ensures modular and reproducible implementation of the framework. The below paragraphs will provide detailed descriptions of all the tools utilized.

3.3.1 OpenCV

The Open Source Computer Vision Library (OpenCV) is employed as the computer vision and image processing library in the proposed ReScene framework. OpenCV processes video frames required for the forgery detection pipeline. In particular, OpenCV is employed to extract frames, select stable frames, preprocess images, and verify the structure between corresponding observations.

Regarding the structural verification branch, OpenCV is applied to calculate the Structural Similarity Index (SSIM) of the reference and revisit images as well as to find and match ORB descriptors in regions where there are structural differences. It gives visual clues for supporting the semantic information that is obtained from the scene graph branch. Consequently, OpenCV is among the main tools for image and video processing tasks within ReScene.

OpenCV also helps to modularize the framework because video processing, feature extraction, and structural similarity are available in one free computer vision library. In that way, it enables the structural verification branch to operate separately from the object detection branch following the hybrid architecture.

3.3.2 Scene Graph Construction and Semantic Analysis

The process of semantic analysis represents the main reasoning part of the proposed ReScene model. This phase aims at turning the observations of the visual nature that have been made during watching the reference and revisit videos into the structured description of the objects and their spatial relations. Rather than detecting the objects separately, the proposed method creates the

graph, where the nodes represent the objects, and the edges represent their spatial relations.

YOLOv8n detector is used for each selected frame to detect the objects existing in the scene. Every detected object is converted into a node of the scene graph. Next, spatial relations between the detected objects are computed based on their positions in the frame. The spatial relations taken into account in the proposed framework include LEFT_OF, RIGHT_OF, ABOVE, BELOW, and NEAR. Thus, the scene graph can be presented as

$$G = (V, E)$$

where V is the set of objects detected and $E$is the set of spatial relations between them. This kind of representation helps the system capture not just the presence of the objects, but also their arrangement.

The scene graphs derived from the reference video are stored in a Scene Memory. The memory contains the repeated appearances of the objects, relations, and contextual associations noted in the reference video. Therefore, the semantic representation is not confined to a single snapshot, but is built on the information gained from the entire video.

The resulting scene representation is then processed via MemoryGCN, which is built on top of a Graph Convolutional Network. The graph convolution process involves exchanging information among adjacent nodes, thereby enabling the object representation to have context information from neighboring nodes. The two-layer GCN adopted in this framework can be expressed as

$$\mathrm{H}^{(l+1)} = \sigma\left(\hat{D}^{-\frac{1}{2}}\hat{A}\hat{D}^{-\frac{1}{2}}H^{(l)}W^{(l)}\right)$$

Thus, the scene-graph construction and semantic-analysis stage enables ReScene to detect changes at the object and relationship levels, including object addition, deletion, replacement, movement, and modification of spatial relationships. It provides the semantic

component of the framework, while the independent structural pathway supplies complementary visual evidence.

3.3.3 YOLOv8 and Pytorch

The YOLOv8 is utilized as the object-detection module in the semantic pathway. The realization of the semantic pathway is done using the lightweight YOLOv8n object detection model.

This model outputs the object detection results, including the class and spatial information for each object. They are further utilized to generate the scene graph.

For instance, suppose there is a person, a car, and a table in a frame. In that case, these objects can be considered as nodes in the graph while the edges will represent their spatial relationships.

The significance of YOLOv8 usage lies in the fact that the semantic pathway needs to comprehend the entities within the scene before making any inference on their relations.

Nevertheless, the object detection process brings a limitation into consideration. This is because the semantic pathway relies on the classes that the detector recognizes. Hence, in case an entity is poorly described by the classes available in the detector, the semantic pathway will have insufficient evidence. This is one of the motivations behind having an independent structural pathway.

PyTorch is the deep-learning framework for the semantic model. It provides such functionality as model training, tensor operations, optimization, and GPU support.

MemoryGCN is trained through Adam optimization with a learning rate of 1×10^{-3} and a batch size of 32. Dropout and early stopping are also applied to prevent overfitting.

3.3.4 PyTorch Geometric and Graph Convolutional Networks

PyTorch Geometric is the framework used for the graph-learning part of ReScene.

In contrast with a traditional image classification algorithm, a graph neural network can explicitly capture the relations between entities. This suits the problem of revisit scene analysis since the identity of

the entity might stay the same but the relations between objects might have changed.

The graph representation of a scene graph looks as follows:

$$G=(V,E)$$

Here, V stands for the objects that were detected and E stands for the relations between them.

A two-layered Graph Convolutional Network is implemented in the methodology proposed. The graph convolution is calculated using the following formula:

$$H^{(l+1)} = \sigma\left(\hat{D}^{-\frac{1}{2}}\hat{A}\hat{D}^{-\frac{1}{2}}H^{(l)}W^{(l)}\right)$$

In this case, $\hat{A} = A + I$is the adjacency matrix with self-loops$\widehat{D}$ is the degree matrix, $W^{(l)}$stands for the trainable weights, and σ is the activation function.

The graph convolution enables passing the information from the neighboring objects. Hence, the model is able to consider the contextual relations instead of looking at each detected object separately.

3.3.5 Scene Graphs and Scene Memory

Scene graphs contain the semantic representation that has been suggested by the proposed system. Each detected object is considered a node, while spatial relationships, like left-of, right-of, above, below, and near, become edges.

The benefit of the representation is that it contains the context of the scene.For instance, one can consider the reference scene where there is a person standing near a vehicle. In the case of a revisitation, the person and the vehicle will be detected as well. However, if the vehicle has moved much since that time, the labels of the objects have not changed, yet the relationship between them has altered.The scene graph is capable of representing this change.The proposed Scene Memory goes even further by considering several frames at once. Rather than looking at each frame individually, the system accumulates object appearances, relationship frequency, and while

the memory graph contains the most common objects and their relations, the relationship analysis focuses on frequent pairs of objects.

3.4 Proposed method

The proposed ReScene method relies on two processing pathways: semantic scene analysis and structural verification [70].

The inputs are a reference video (V_r) and a query/revisit video (V_q). Representative frames are selected from each video at regular intervals [32]:

$$F=\{f_1, f_2, \ldots, f_N\}$$

(N) is the number of extracted frames.

Representative frames are analyzed by the semantic and structural pathways.

For the semantic pathway, object detection, scene graph generation, and storing data into the Scene Memory take place. MemoryGCN analyzes graph data to check the consistency of semantic relations [18].

At the same time, in the structural pathway, stable frames are selected, and SSIM comparison of reference and query frame is performed. Areas with a low level of similarity are further examined through ORB feature matching [42].

The outputs from both pathways are fused using adaptive confidence fusion to get the score used for making the tampering decision [46].

3.4.1 Video Input and Frame Extraction

The initial step in the process suggested by the methodology is the creation of the reference and the query videos [70].

The reference video is an initial image of the scene, whereas the query video is the same scene, observed at a different point in time. The latter may or may not include one or several modifications [42].

As direct processing of all frames in a video requires significant computational resources and provides redundant information, representative frames are extracted in regular intervals [32].

The set of extracted frames is represented by:

$$F=\{f_1, f_2, \ldots, f_N\}$$

The frames chosen act as the input data for the two principal branches of the methodology.

The semantic branch of the methodology uses the frames for object identification and scene graph creation [22]. The structural branch compares corresponding frames based on their visual structure [46].

For this purpose, stable frames are selected from both the reference and query sequences, which eliminates the impact of the natural motion on the result [42].

The process can be represented as:

$$\text{Video} \rightarrow \text{Frame Extraction} \rightarrow \text{Representative Frames} \rightarrow \text{Semantic/Structural Processing}$$

3.4.2 Semantic Scene Understanding

Once the representative frames are extracted, YOLOv8n is used to detect objects [22].

Every detected object becomes a node of the scene graph. The relation between objects based on their spatial location is used to create edges [22].

For instance,

$$\text{Person} \rightarrow \text{Near} \rightarrow \text{Car}$$

can show the relation between the two detected objects.

The whole graph is presented as:

$$G = (V, E)$$

where (V) are objects and (E) are relations.

This approach allows for getting information that goes beyond object detection. It shows how objects are located in the scene.

Thus, the semantic understanding is able to detect various events like adding/removing, replacing, and moving objects [22].

For instance, an object can be moved from one side of the scene to another. While the class of the object will remain the same, its relations with other objects can change. This information will be included into the scene graph [18].

The constructed graphs based on several frames are used to form the Scene Memory.

3.4.3 Scene Memory and Semantic Inconsistency Detection

The Scene Memory keeps the information received from the reference video in several consecutive frames.

The memory includes the statistical data concerning:

- The appearances of objects
- The frequency of relations
- Co-occurrence of objects
- The context of appearance

Thus, the system is able to create a representation of the usual observation in the reference scene.

This graph information is further transferred to the MemoryGCN. The GCN transfers the information between the nodes, helping the model to learn the dependencies in the context.

Due to the shortage of publicly accessible datasets with semantic scene-graph forgery, the research methodology generates synthetic forged graphs by modifying the authentic graphs of scenes. Five ways of the modification are used: object replacement, relation corruption, random object addition, edge deletion, and spatial perturbation.

The semantic confidence score generated by the MemoryGCN is equal to:

$$S_{sem} \in [0, 1]$$

The higher the value of this metric is, the more reliable is the inconsistency between the semantic structures of the reference and revisit scenes.

It helps the model to detect the inconsistencies even if the changes have occurred in the relations between the objects.

3.4.4 Structural Verification

The structural verification module works separately from the semantic branch [46].

This separation is necessary because the object detector might not detect all objects and might not be able to offer sufficient semantic

clues for some manipulations [65].

The structural branch starts with choosing the stable frames from the reference and the query videos. These frames are compared with SSIM [46].

The regions that have low structural similarity are then analyzed with ORB feature matching.

This process can be expressed as:

$$\text{Reference/Query Frames} \rightarrow \text{SSIM} \rightarrow \text{Suspicious Regions} \rightarrow \text{ORB} \rightarrow S_{\text{struct}}$$

The structural confidence is scaled to:

$$S_{\text{sem}} \in [0, 1]$$

where higher values correspond to higher evidence of structural difference.

This module is able to find local visual changes in spite of having no change in the semantic representation [46].

For example, an object can remain classified as the same object both before and after manipulation. But visual features of the object can differ enough to make the SSIM and ORB detection modules recognize suspicious regions.

3.4.5 Adaptive Fusion and Final Decision

When semantic and structural branches finish analyzing, they combine their confidence scores.

The score for the semantic branch is denoted as: S_{sem}

and for the structural branch as: S_{struct}

$$S_{struct}$$

The adaptive weight is defined as:

$$\text{Reference/Query } \alpha = \frac{S_{\text{sem}}}{S_{\text{sem}} + S_{\text{struct}} + \epsilon}$$

and the confidence for the final decision is:

$$S_{\text{final}} = \alpha S_{\text{sem}} + (1 - \alpha) S_{\text{struct}}$$

The adaptive fusion method is shown in Figure 1. There is a combination of semantic and structural scores prior to the final decision.

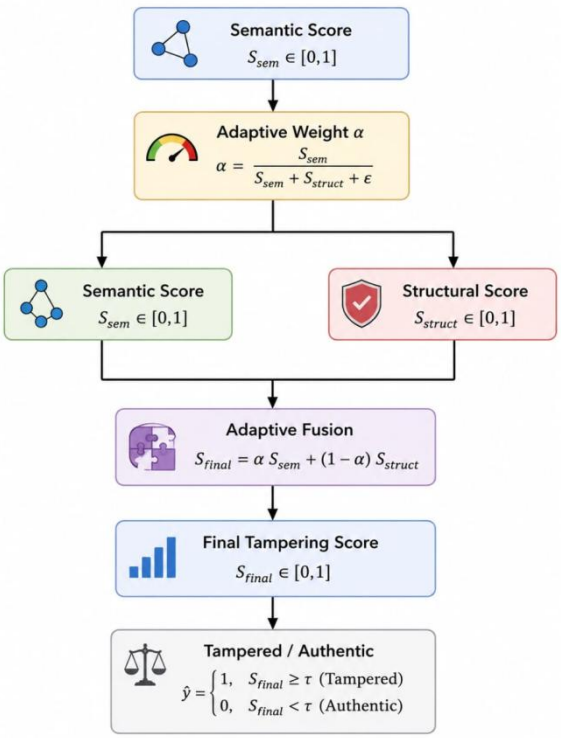


Figure 1. Adaptive confidence fusion approach combining semantic and structural confidence measures to make the final prediction for video manipulation.

The point of using adaptive fusion is to enable the final decision to take into account the relative importance of both pieces of evidence.

The rule for the fusion diagram is

$$\hat{y} = \begin{cases} 1, & S_{\text{final}} \geq \tau \text{ (Tampered)} \\ 0, & S_{\text{final}} < \tau \text{ (Authentic)} \end{cases}$$

If there is an object addition or removal, the semantic branch may analyze the modified object and its relations, whereas the structural branch may find the local visual change corresponding to it.

For object displacement, the semantic branch finds the modified spatial relation, and the structural branch finds visual feature-mismatch evidence.

In case of unknown objects, the semantic evidence will be poor, but the structural branch may find the visual inconsistency evidence, which does not require object classification.

3.5 Work Flow of the System

The complete workflow of the proposed ReScene system is desribed in the figure 2.

The whole process starts by taking representative frames from reference video and query video. The two processes start separately after which their results are integrated [70].

For the semantic branch, YOLOv8n is utilized to detect objects [22]. Objects are transformed into scenes graphs and the multiple information frames are accumulated into the Scene Memory. MemoryGCN is applied on the graph representation to detect semantic discrepancies[18].

For the structural branch, stable correspondences are selected. Regions containing structural discrepancies are detected by SSIM, while ORB feature matching acts as additional support for that detection[46].

Finally, the results from both branches are normalized and fused together through adaptive confidence fusion. The (S_{final}) is used to indicate the tampering confidence.

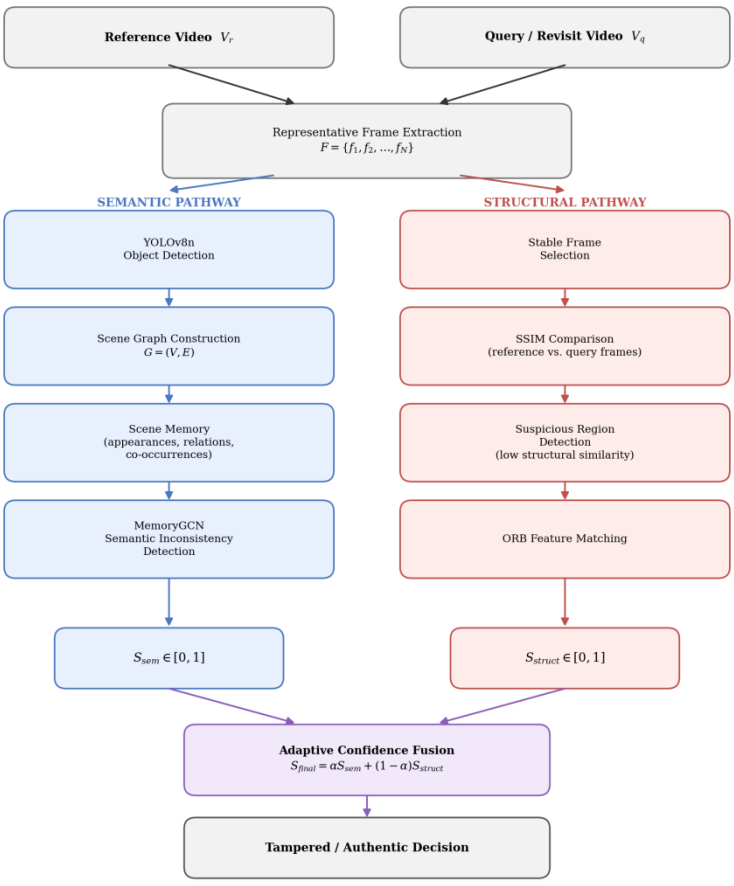


Figure 2. Work flow of the System

CHAPTER 4

RESULTS AND DISCUSSION

4.1 Experimental Setup

The a fore mentioned ReScene method was designed and examined to determine the potential of this method in detecting any kind of semantic or structural inconsistency between the reference video and either of the query/revisit videos taken from the same physical environment. The traditional video forgery detection techniques mainly rely on visual artifacts, temporal inconsistency, frequency domain analysis, or appearance learning features. These methods have proven to be quite effective in dealing with certain classes of manipulated videos, but the performance of these techniques may get compromised in situations where the manipulation does not generate strong visual artifacts [8], [10], [35], [64].

The experimental procedure starts by selecting two videos of the same physical space. One video will be taken as the reference video, whereas the other will be referred to as the query video or revisit video. The query video will depict either an actual revisiting of the same physical space or a tampered video having scene-level changes. Frames from both videos are selected to save computational effort. Frame-level and temporal analysis methods have previously been utilized in video forgery detection owing to the usefulness of frame inconsistencies as evidence of forgery [32], [36], [46], [50].

The selected frames are processed using two separate processing paths. The first path is the semantic processing path wherein the objects in the image are detected using YOLOv8n and a graph is constructed based on the objects and their spatial relationships. Information collected from different frames is collected in the Scene Memory. The MemoryGCN model then processes the graph to find inconsistencies in semantic structure. Graph representations are helpful in modeling the relationships between different visual entities

since they carry more information than independent object classification [22].

The second branch carries out the structural validation. Comparison of stable frames between reference and query video is performed through Structural Similarity Index Measure (SSIM). Regions with poor structural similarity are analyzed using ORB feature matching. Visual and inter-frame inconsistencies have been identified based on similarity and feature-based approaches for video tamper detection [42], [50].

Outputs of both branches are normalized and fused using the adaptive confidence fusion scheme. This value S_{final} is the tampering confidence of the system. This means that the experiment will test not only how ReScene is able to detect manipulated videos, but also whether the two branches are useful for different kinds of manipulations.

Table 1. Experimental Components

Component	**Technology/Method**	**Purpose**
Video processing	OpenCV	Frame extraction and image processing
Object detection	YOLOv8n	Detection of semantic objects
Scene representation	Scene Graph	Representation of objects and relations
Semantic memory	Scene Memory	Accumulation of information across frames
Graph reasoning	MemoryGCN	Detection of semantic inconsistencies
Structural comparison	SSIM	Detection of structural differences
Local verification	ORB	Feature-based verification of suspicious regions

Fusion	Adaptive Confidence Fusion	Combination of semantic and structural evidence
Deep learning framework	PyTorch	Model implementation and training
Graph learning framework	PyTorch Geometric	Graph-based learning

The design ensures that there is always a separation between the training and the evaluation phase. The synthetic semantic manipulation data sets are generated based on realistic scene images used to train the semantic detection framework. Evaluation is done using reference and query video sets that contain realistic and manipulated data sets.

4.1.1 Dataset and Benchmark Construction

One of the main issues when assessing the proposed framework was the absence of an open-source large dataset for the semantic part of ReScene manipulations. Indeed, video forgery datasets greatly assisted in forging detection technique development; however, many of the current benchmarks concentrate mainly on facial manipulation, synthetic media, or traditional video-level forgery [5], [15]. Thus, a special synthetic benchmark was created for testing the semantic aspect of ReScene.

The initial step in creating the benchmark is processing the authentic video frames with the proposed scene graph generation algorithm. The obtained scene graphs depict the objects and relations seen in the authentic scene. Next, forged sample graphs are generated via semantic manipulations of the authentic graphs.

The following five kinds of semantic manipulation were chosen:

1. Object replacement
2. Relation corruption
3. Random object addition
4. Edge deletion
5. Spatial perturbation

These manipulations were chosen due to their semantic inconsistency nature. Object replacement replaces an entity identity, random object addition adds a new entity, edge deletion deletes an existing relation, relation corruption changes the context relationship between the entities, and spatial perturbation changes the location of an object and thus several relationships.

This is because the generated dataset has an equal number of genuine and fake entries in an effort to mitigate the effects of class imbalance. The entries were randomized and partitioned in an 80:20 ratio using stratified sampling.

4.1.2 Revisit Video Test Configuration

The full ReScene approach was tested independently of the semantic model training dataset. In the revisit-video testing scheme, the scenes consist of a reference video together with genuine and fake versions of the revisit video. Reference and query videos correspond to the observation of the same scene whereas the forged queries involve scene manipulation.

The testing includes scenarios such as object insertion, object removal, object substitution, and object movement. It is crucial to note that the full ReScene is supposed to function on actual pairs of videos as opposed to individually manipulated scene graphs.

It is also important that reference and revisit videos allow the ReScene system to test for consistency and not just presence of artifacts in a single frame.

4.1.3 Experimental Environment

Implementation of the proposed ReScene approach was done with the use of Python, PyTorch, PyTorch Geometric, OpenCV, and YOLOv8n. YOLOv8n was used for object detection and scene graph creation. OpenCV was used for frames extraction, stabilization, SSIM-based structural similarity, and ORB features extraction. Graph-based semantic model was implemented with the use of the Graph Convolutional Network architecture available in PyTorch Geometric [8], [20].

MemoryGCN was trained with Adam optimizer, the learning rate being 1×10^{-3}, and batch size being 32.

4.2 Semantic Branch Results

The responsibility of the semantic part is to detect the inconsistencies between the objects and relations existing in both the reference and the query scenes. Scene graphs offer an organizational structure where the detected objects are denoted by nodes and the relation between them is denoted by edges. Scene graph based representations have been extensively studied for expressing relations between objects [6], [7], [17].

In ReScene, the semantic representation is used for forensic purposes instead of using it for standard scene understanding. The aim here is to detect whether the semantic structure of the query scene is consistent with that of the reference scene.

4.2.1 Object Detection Results

The YOLOv8n model is employed in detecting objects in the representative frames. In doing this, it gives the object category and location that are necessary for the generation of the scene graph. Thus, the object detector is the first perception module in the suggested system [20].

The accuracy of the object detection process will affect the accuracy of the scene graph generated. If the objects are reliably detected, then the relations among them can be formed and compared across observations. Otherwise, failure to do so will lead to an inaccurate scene graph.

Thus, the object detection step itself does not constitute the forgery detection procedure. It only provides the semantically important information necessary for the next MemoryGCN reasoning step.

4.2.2 Scene Graph Generation Results

Once objects are detected, every one of the detected objects is mapped to a node in the graph. Spatial relations between the objects including left-of, right-of, above, below, and near are represented using graph edges. The scene graph is expressed as follows:

$$G = (V, E)$$

where V denotes the detected objects while E denotes the relations among the detected objects.

This mapping provides more than information about the objects themselves. For example,

$$\text{Person} \rightarrow \text{Near} \rightarrow \text{Car}$$

denotes a relation between two objects. If one or both of these objects is still there but the spatial relationship between them has changed, then the graph structure has also changed.

This is very useful when an object is displaced. This means that the object keeps the same category before and after manipulation; therefore, a regular object detector does not provide any different label for the object. However, the relationships between the object and other objects might have changed.

4.2.3 Scene Memory Analysis

The Scene Memory stores semantic information derived from several frames of the reference video. The aim of such a memory representation is to create a persistent representation of the scene without using any individual frame.

The Scene Memory stores object occurrences, relationship occurrences, and contextual co-occurrences in the video. Thus, any object which occurs several times in various frames contributes towards creating a more persistent representation of the reference scene.

In the left-hand side of the figure 3, the relationship graph learned by our model is shown, with the size of nodes being proportional to the object frequency and the width of edges being proportional to the relationship frequency. In the right-hand side of the figure 3, frequent object pairs are shown.

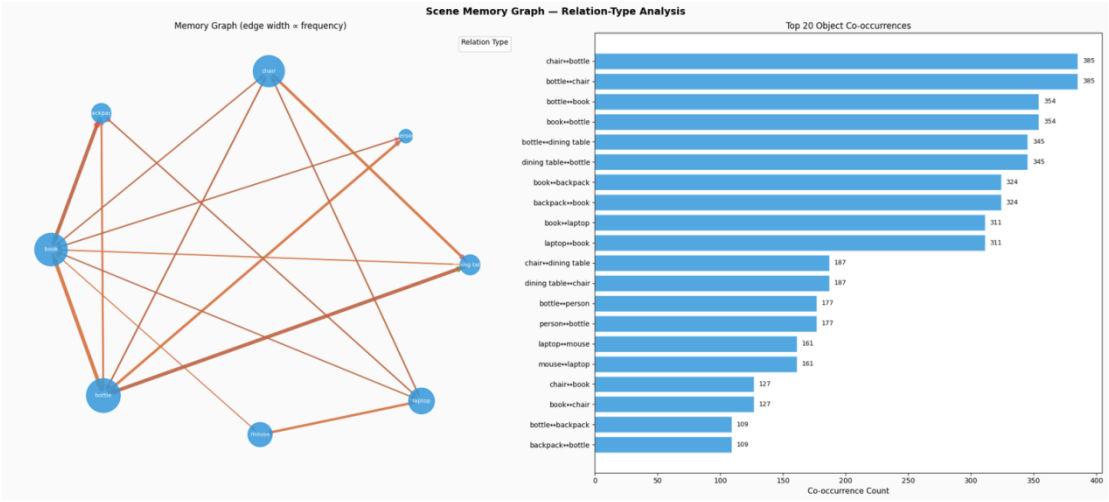


Figure 3. Illustration of Scene Memory representation that would be employed in the proposed ReScene system.

The Scene Memory framework works well in revisit recognition because one frame could give incomplete data because of occlusion or absence of the object. With many frames, the Scene Memory framework could make a more durable representation of the scene.

4.2.4 MemoryGCN Semantic Detection Results

The constructed scene graphs are fed into the MemoryGCN network. The fundamental mechanism of the graph convolution process involves spreading out context information among neighboring nodes in order to allow the network to learn dependencies between objects and their relations [8].

The confidence of semantic inconsistency is defined as:

$$S_{\text{sem}} \in [0, 1]$$

with higher values implying greater evidence for semantic inconsistency.

The semantic network has been trained using graph forgeries artificially created for the purpose. The employment of such techniques as object replacement, relation corruption, object insertion, edge removal, and spatial perturbation provides the network with multiple types of graph-level inconsistencies.

As a result, the semantic network should be especially useful for manipulating object identity or spatial relations.

4.3 Structural Verification Results

The structural verification module works independently from the object detector. This is necessary due to the fact that sometimes there will be insufficient data to perform the manipulation detection based on semantics. The same category in the object detector may apply to an object regardless of its manipulation.

The structural module then compares the visual similarity between reference and query frame images using SSIM and ORB.

4.3.1 Stable Frame Selection

Stable frames are chosen from the reference and query sequences prior to structure analysis. The goal of this step is to minimize the effect of natural motions and unstable frames.

If no stable frame selection had been carried out, it would be possible for natural camera movements or other factors to cause differences in structure which are not a result of any manipulation.

4.3.2 SSIM-Based Structural Analysis

SSIM is employed for calculating structural similarity in reference and query frames. The SSIM measure uses local luminance, variance, and covariance attributes to calculate the structural similarity [9].

The SSIM formula utilized in this method is as follows:

$$\mathrm{SSIM}(x,y)=\frac{(2\mu_x\mu_y+C_1)(2\sigma_{xy}+C_2)}{(\mu_x^2+\mu_y^2+C_1)(\sigma_x^2+\sigma_y^2+C_2)}$$

Where μ_x and μ_y stand for local mean values, σ_x^2 and σ_y^2 are local variance values, and σ_{xy} is a local covariance value [9].

Areas with low structural similarity are regarded as suspicious regions and handed over to ORB verification stage.

4.3.3 ORB Feature Matching Results

ORB serves to determine local feature correspondence between suspicious regions. ORB is a fast algorithm to detect and describe features and was presented as a replacement to the expensive feature description algorithms [10].

In ReScene, ORB offers additional support for determining if there is feature correspondence between the reference and query regions. Reduction in feature correspondence within the suspicious region means that structural changes have occurred in the region.

4.3.4 Structural Confidence Analysis

Normalized confidence measure is computed by the structure-based branch:

$$S_{sem} \in [0, 1]$$

The larger this number is, the stronger the structural inconsistency is detected.

The resulting confidence measure is then used at the adaptive fusion stage. This independent structure-based branch enables the ReScene method to detect visual changes in local regions independently of changes in the semantics.

4.4 Case Study Analysis

The practical behavior of ReScene was studied through various revisit-scene manipulation cases. The IEEE paper that was uploaded indicates that there are three main categories of cases in a case study: object adding/removing, object moving, and unknown/modified objects. The following are examples of how the two branches work together.

Table 2. Case Study Analysis

Case	Manipulation	Semantic Evidence	Structural Evidence
1	Object addition/removal	Object/relation change	Local structural change
2	Object displacement	Spatial relation change	Feature mismatch
3	Unknown/altered object	Limited detector evidence	SSIM/ORB inconsistency

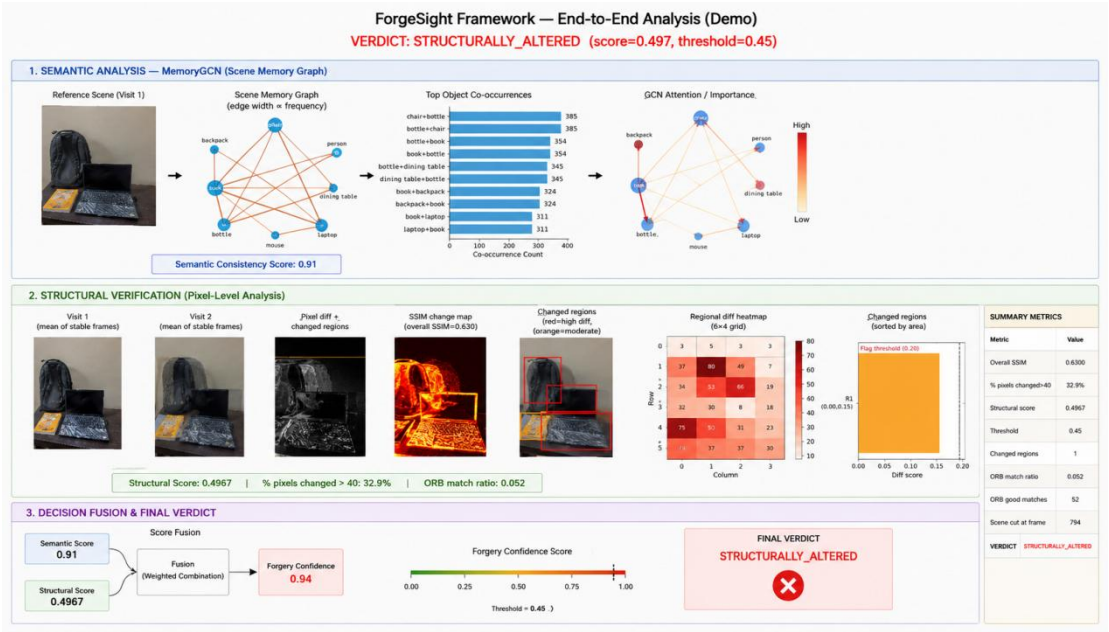


Figure 4. Representative qualitative results of the proposed ReScene framework

Figure 4 represents the qualitative results in the proposed Rescene framework . It shows the authentic and forged revisit scenes together with the corresponding semantic and structural analysis.

4.4.1 Object Addition and Removal

The process of adding or removing objects from the input video directly affects the semantic structure of the scene. If an object is added to the input video, then it can be possible that the object detector will detect an object that is not present in the reference scene. In this way, there will be a difference between the scenes in terms of their graphs.

Likewise, if an object is deleted, then the nodes, as well as relations related to them, may not be present in the graph of the query video.

In this scenario, the use of semantic pathway can be very useful as there will be a difference both in the objects set and in their relations structure, while the structural pathway will provide evidence about the visual difference in the corresponding image region..

4.4.2 Object Displacement

In the case of object displacement, there will be a higher difficulty in manipulating the image since the object might still belong to the same class.

If the chair is displaced to the other side of the table, YOLOv8n will detect a chair in both the reference and query video sequences. This means that classifying objects alone is not enough to prove manipulation.

But the relations of the chair relative to other objects have now changed. The edges in the semantic graph have now changed, hence indicating spatial manipulation. MemoryGCN can utilize such relational differences to detect semantic inconsistencies.

There is further evidence from the structural branch in that the position of the object and the corresponding background will be different. There will also be local feature verification using ORB feature matching.

4.4.3 Object Replacement

Replacement of object transformation involves the substitution of one object with another or one existing entity with another semantic category. The alteration can influence the node representation of the graph.

In case an object category within the reference scene gets replaced by another category within the query video, there is a difference in the node feature. Even the relations linked with the node could get altered.

Thus, the semantic branch can be used as an evidence of the transformation. The structural branch will serve as further visual evidence of the transformation within the region where it has taken place.

4.4.4 Unknown or Altered Objects

A significant problem in the semantics sub-module is when there is poor representation of the object in the vocabulary of the detector. Object detectors are designed to run against specific categories of

objects, which means that in cases where an object is new or altered, there is insufficient semantic evidence.

For such situations, the structural sub-module is an alternative because it works independently of any object recognition.

The scenario above presents a good example of the rationale behind the design of the hybrid approach.

4.5 Accuracy And Performance Metrics

The performance of the proposed ReScene framework was analyzed with standard classification metrics in order to evaluate the capability of the framework for distinguishing authentic from forged revisit videos. The following metrics were chosen due to their complementary nature for classification performance analysis of the proposed system:

Accuracy is the percentage of correctly classified samples over all the examined samples and can be calculated using the formula:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

where TP, TN, FP, and FN are true positives, true negatives, false positives, and false negatives, correspondingly.

Precision is the percentage of samples classified as forged which are really forged and can be calculated as:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Recall is the percentage of real forged samples which are correctly recognized by the proposed framework and can be calculated as:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

F1-score combines precision and recall and can be calculated using the formula:

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

As a result of the reported evaluation of the complete ReScene framework there were obtained 0.77 accuracy, 0.78 precision, 0.75 recall, and 0.76 F1-score for the customized revisited-scene dataset.

Table 3. Performance of ReScene on the custom revisited-scene dataset.

Method	Accuracy	Precision	Recall	F1-Score
ReScene	0.77	0.78	0.75	0.76

The accuracy value of 0.77 shows that the proposed approach successfully classified 77% of the tested samples. The precision value of 0.78 shows that the majority of the samples classified as forged actually were forged. The recall value of 0.75 means that the system managed to identify 75% of the forged samples within the tested data set. The F1-score value of 0.76 is the combined metric of both precision and recall.

The achieved performance was provided by the whole hybrid pipeline and not one detection part. In the semantic branch, we get information about objects and relations between them, while in the structural branch, we get information about visual differences between corresponding areas.

Semantic representation is particularly useful in cases when manipulation is related to object addition, deletion, substitution, or movement. With scene graph based representation, we can explicitly represent relationships between the objects, thus detecting some changes which can not be found using only object categories [6], [7]. MemoryGCN helps to process the information about relations through graph based learning [8].

structural branch provides complementary evidence. SSIM identifies regions with reduced structural similarity, while ORB provides local feature correspondence for suspicious regions [9], [10]. This is useful when an object remains within the same semantic category even though its visual appearance or local structure has changed.

The final decision is based on the adaptive combination of the two confidence scores:

$$\mathrm{S}_{\mathrm{final}} = \alpha S_{\mathrm{sem}} + (1 - \alpha) S_{\mathrm{struct}}$$

This allows the relative contribution of semantic and structural evidence to vary according to the information available for a particular scene.

For example, when an additional object is inserted into the query scene, the semantic branch can identify a difference in the object and relationship structure. In contrast, when an object retains its semantic identity but undergoes a local visual modification, the structural branch can provide additional evidence through SSIM and ORB.

The performance results should therefore be interpreted as evidence of the operation of the complete hybrid framework. Separate numerical performance values for semantic-only and structural-only configurations should only be reported if those experiments were independently conducted.

4.6 Discussion of Results

4.6.1 Semantic Branch Analysis

The semantic module of the introduced ReScene system deals with the detection of discrepancies in the semantic composition and the spatial relationship of the revisited scene. YOLOv8n extracts object instances in the reference and revisit video sequences, and they become nodes of scene graphs while spatial relationships between objects are encoded as edges of the scene graphs. Scene Memory keeps the information about object instances and spatial relationships, whereas MemoryGCN analyzes the scene graph to detect semantic inconsistencies.

The semantic module becomes especially effective in case of manipulations when the appearance of an object does not change, but its relationships with other objects become different. For instance, displacing an object changes spatial relationships of the type LEFT_OF, RIGHT_OF, ABOVE, BELOW, NEAR without altering the object category.

4.6.2 Structural Verification Analysis

The structural verification branch independently analyses the visual characteristics of both the reference and revisit frames. Structural discrepancies in stable frames are determined by the use of SSIM, while local visual correspondences are verified by the use of ORB feature matching. Contrary to the semantic branch, the process does not require any categories of objects detected.

This branch is useful when the manipulation cannot be detected by the semantic detector. The unknown or altered objects may contain little or no semantic discrepancy but the discrepancy in the structure and local features may point out that there is manipulation in the image.

4.6.3 Combined Performance and Case Study Analysis

Complementary proof is provided by the semantic and structure branches. The semantic branch can show what has been modified in terms of objects and their interactions, whereas the structure branch can highlight where there are differences visually using SSIM and ORB. By using the adaptive confidence fusion method, the final tampering confidence will be acquired.

The reported result on the custom revisited scene dataset is accuracy 0.77, precision 0.78, recall 0.75, and F1 score 0.76. Several examples that include adding, deleting, moving objects, and unknown or modified objects show how both branches provide proof to the final proof process.

4.6.4 Overall Interpretation and Limitations

The results favor the application of a hybrid semantic and structural method for forgery detection in videos that rely on revisit detection. Semantic module uses information from Scene Graph and Scene Memory, while the structural module uses information from SSIM and ORB. Such a combination makes it possible to deal with various kinds of scene manipulations.

The results also indicate several drawbacks. For the semantic module, the quality of the object detection is important, whereas the semantic

training dataset was artificially produced because there are no appropriate large-scale datasets available in the public domain. Structural verification can also be impacted by the differences between reference and revisit images.

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

One of the crucial problems in video forensics is the problem of detecting any form of tampering in cases where the same physical setting is viewed repeatedly. Traditional methods used to detect forgery in videos mostly concentrate on face artifacts, pixel discrepancies, temporal discrepancies, or appearance based discrepancies. These techniques may fail to capture any form of tampering at the scene level such as addition of an object, deletion of an object, replacing an object, displacement of an object, etc. In the proposed technique, ReScene: A Hybrid Semantic and Structural Framework for Revisit Based Video Forgery Detection, the scene level discrepancy is identified using the reference and revisit videos.

The approach involves the integration of semantic understanding of the scene with its structural visual verification. Semantic stream leverages the YOLOv8n model for detecting objects present in both the reference and the revisit video and encodes them as nodes in a scene graph. Spatial relationships between the objects are encoded as graph edges enabling reasoning about the structure of the scene instead of analyzing individual objects. The scene memory retains information about objects and relations between them in the reference video, while MemoryGCN processes the graph encoding for detecting semantic inconsistencies in the revisit scene.

The structural branch serves as another independent source of visual proof. Structural Similarity Index Measure (SSIM) is used for comparing stable frames in the reference and revisit videos, and local features are matched with ORB in order to confirm the findings. The structural branch works well when it is impossible to recognize a manipulated object by means of the semantic detector. Thus, the structural analysis adds value to the semantic reasoning by taking

into consideration the visual changes which are not reflected in the scene graph.

As for the experimental results, they prove the effectiveness of the proposed method. Using the custom revisited-scene dataset, the accuracy was 0.77, precision was 0.78, recall was 0.75, and the F1-score was 0.76. There were various cases of scene-level manipulation taken into account: adding an object, removing an object, displacing an object, and the unknown or changed object.

The adaptive confidence fusion stage further fuses the outputs from both the semantic and the structural branches to determine the final tampering confidence. Semantic branch gives information concerning the change in object identity and their relative spatial relations whereas the structural branch gives information about changes based on visual resemblance and matching of local features. This reduces reliance on a single type of evidence and gives a more comprehensive view of the consistency of the scene. This model creates a link between semantic reasoning and traditional image verification techniques in a video forgery detection process.

The proposed technique also shows the importance of taking into consideration the physical situation of the scene while carrying out forgery analysis in video content. The manipulation is not necessarily meant to change the appearance of the objects to make it detectable. Objects may remain visually similar but have a different relationship with other objects around them. Additionally, an object that cannot be classified semantically due to being unusual or changed may lead to inconsistency in the structural analysis of the scene.

In general, the suggested framework presents an organized scheme for revisit-based video forgery detection by using computer vision and deep learning techniques that are generally available. While there are certain drawbacks to the current design of the system due to object detection, limited data, natural variations in the scenes, and computing process issues, the findings indicate that the combination of semantic persistence and structural validation can be considered a

promising path for video forensic investigation on the level of the scene.

5.2 Limitations of the proposed framework

While promising results have been obtained using the proposed framework, there are certain limitations associated with them that need to be taken into consideration to maintain the integrity of the scope of the research.

- **Accuracy of Object Detection**

The proposed semantic branch relies on the YOLOv8n model for object detection. Improper object detection, missing detection or incorrect bounding boxes may influence scene graph generation, which, in turn, will influence the semantic confidence provided by the MemoryGCN.

- **Limited Semantic Vocabulary**

The semantic branch relies on the ability of the detection model to detect certain object categories. Objects that have not been seen by the detection model or have undergone considerable changes may not be appropriately represented within the scene graph, which will limit the semantic evidence available to detect a manipulation.

- **Reliance on Synthetic Data**

As a result of the lack of appropriate semantic manipulation data, the training process for the semantic branch relied on synthetic scene graph manipulations.

- **Sensitivity to Variations in the Natural Scene**

It is possible that the variations observed in the reference and revisit videos could have been caused by illumination, view, camera angle, shadow, or variation in the environment surrounding the object in the videos. This variation could have created structural or semantic differences in the absence of any forgery, which makes it difficult to distinguish natural variation from forgery.

- **Limitations of Structural Comparison**

Although SSIM and ORB provide important evidence based on the structure and features of the frame, they can be affected by variations in view, illumination, scale, and appearance of the local features. The

structural difference between two frames cannot always be used as evidence of forgery.

- **Dependent on Stable Frame Selection**

Selection of appropriate frames is crucial to the success of the structural verification process since improper selection of frames can negatively impact the performance of SSIM and ORB correspondences.

5.3 Future scope

In terms of the future direction of video forgery detection based on revisits, there are several research directions with respect to the proposed ReScene framework.

- Large-Scale Real-World Revisit Dataset

One can collect a large-scale dataset of revisit videos that consist of authentic ones as well as manipulated ones under various environments, viewpoints, recording settings, and types of manipulation.

- Open-Vocabulary Semantic Detection

Open-vocabulary detection methods can be introduced to detect various types of objects, even those which were not seen in training data, which will contribute to better semantic detection of unknown objects or objects which were altered.

- Improved Scene Memory and Graph Reasoning

Scene Memory can be enhanced by enriching information on objects, relationships, and temporal aspects, and graph neural network models can be applied to improve object changes and relationships' detection.

- Temporal Scene Reasoning

In the future, one can consider introducing object tracking and temporal scene reasoning through full video sequence, so that real object movement from forgery, such as displacement or insertion/removal of objects, can be distinguished.

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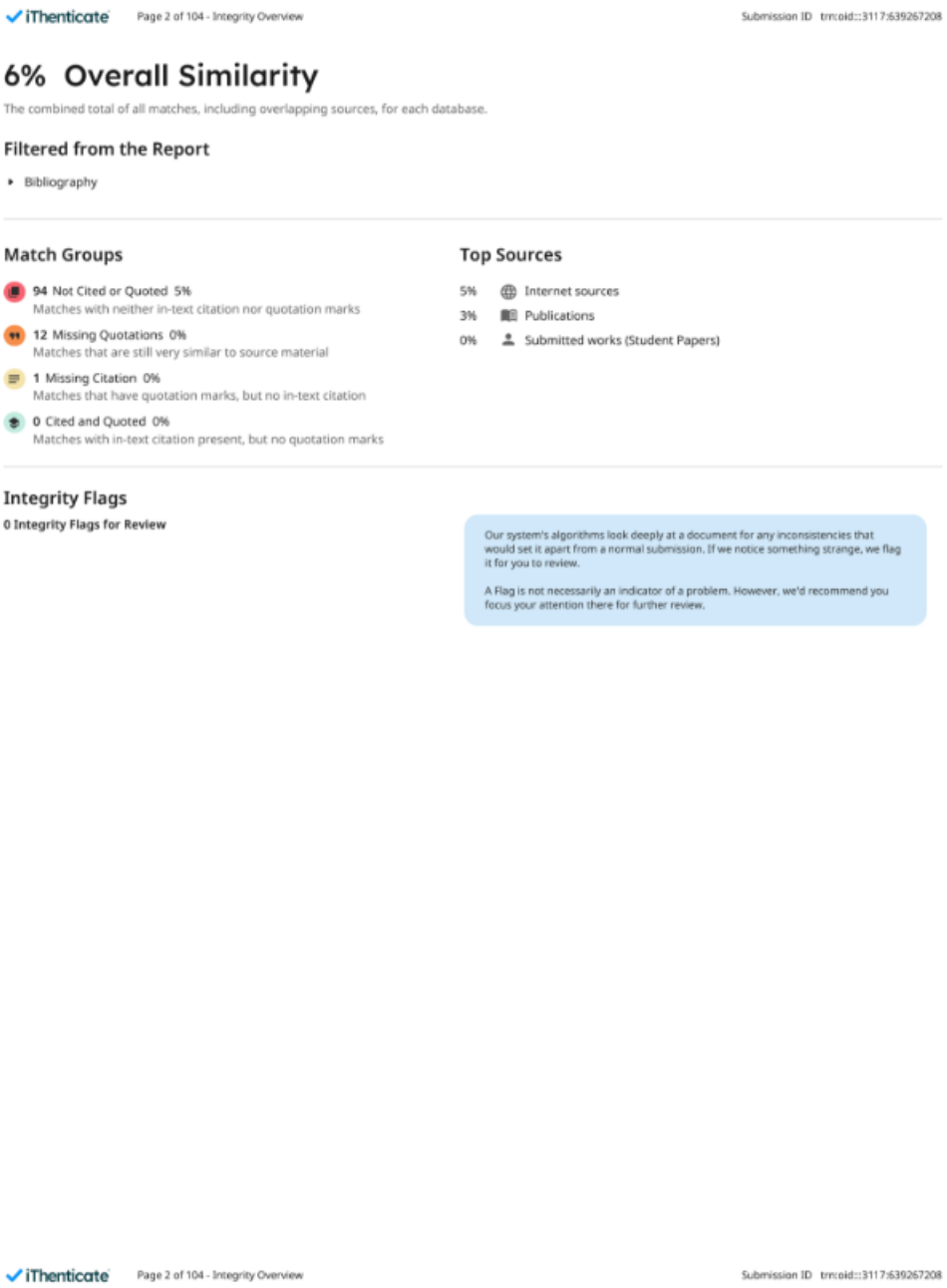
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